Abstract

This report critically analyses Agile Project Management and Privacy by Design through the FarmaY ERP case study. The first section evaluates the use of Agile Scrum in a dynamic and regulated environment, assessing its suitability, implementation approach, and associated practices. While Agile enabled iterative delivery, stakeholder collaboration, and improved responsiveness, its effectiveness was constrained by limitations in role execution, team autonomy, and governance. The second section examines privacy and data protection issues, focusing on access control, data integrity, and challenges of implementing Privacy by Design within the enterprise systems. The findings highlight that privacy considerations were addressed reactively rather than proactively. Overall, the study demonstrates that successful outcomes depend on disciplined Agile execution and early integration of privacy and governance mechanisms.

Acknowledgement

As I bring this project into completion, I am sincerely appreciative for the support, direction, and encouragement I have received throughout this academic journey. From the formulation of concepts to the development of this proposal, I have been fortunate to benefit from the support of many exceptional individuals whose contributions have been important.

First and foremost, I would want to convey my profound gratitude to my lecturers at ESU, Kandy, whose constant dedication, competence, and assistance have substantially expanded my comprehension of the subject matter. I am particularly thankful to Ms. Kaushalya Wijayarathne, whose mentoring has been crucial in assuring the quality of my project.

I would also like to offer my appreciation to my classmates and peers, whose participation, constructive criticism, and encouragement have inspired me to develop my ideas and confront obstacles with confidence. Your support has been a tremendous source of inspiration during this journey. I am extremely grateful to my family, whose constant conviction in my talents and ongoing encouragement have given me with the strength and confidence to pursue my academic aspirations. Your affection and backing have been the cornerstone of all my triumphs.

Finally, I would like to acknowledge London Metropolitan University for providing the curriculum and modules that have guided my study, and ESU, Kandy, for offering the facilities, resources, and academic environment necessary to produce this project. I also convey my thanks to everyone who has contributed, directly or indirectly, to my intellectual and personal development during this endeavor.

Table of Contents

1. Introduction 6
2. Section 01 6
2.1 A Summary of the Case Study 6
2.2 Agile/SCRUM Software Development 7
2.3 Suitability of Agile/SCRUM for the Project 7
2.4 Analysis of the Approach taken by the Project Team 8
2.5 Good Practices 9
2.6 Bad Practices 11
2.7 Lessons Learnt 12
2.8 Key Findings 13
3. Section 02 13
3.1 Privacy by Design 13
3.2 Overview of Issues Raised in the Case Study 14
3.3 Challenges of Implementing Privacy by Design 15
3.4 Key Findings 15
4. Conclusion 16
References 17

Critical Analysis of Agile Project Management and Privacy by Design: A Case Study of ERP Development in the Pharmaceutical Industry.

1. Introduction

This research critically evaluates Agile Project Management and Privacy by Design using the FarmaY ERP case study. The first section assesses the adoption of Agile Scrum, analyzing its applicability, implementation approach, and the associated good and bad practices, followed by significant lessons and discoveries. The second portion focuses on Privacy by Design, providing an outline of its principles, assessing privacy-related concerns within the case, and exploring the obstacles of implementation in corporate systems. Overall, the paper intends to analyze how good project management methods and embedded privacy measures contribute to successful, secure, and compliant system development.

2. Section 01

2.1 A Summary of the Case Study

This case study investigates the deployment of an Agile Scrum-based IT project at FarmaY, a pharmaceutical company creating an Enterprise Resource Planning (ERP) inventory management module to prevent losses from outdated pharmaceuticals. The project implemented in a highly regulated and dynamic environment, where inventory circumstances and compliance requirements regularly changed, resulting in evolving and initially unclear system requirements.

To address these challenges, the business implemented the Scrum framework, using iterative sprints and constant stakeholder interaction to deliver functional increments iteratively. Agile approaches are extensively employed in uncertain contexts due to their flexibility and capacity to meet changing requirements (Rigby, et al., 2018). They also promote early delivery and enhanced response, improving corporate value (Project Management Institute, 2021; Digital.ai, 2023).

2.2 Agile/SCRUM Software Development

Agile software development is an iterative and incremental approach designed to manage projects in environments defined by unpredictability and evolving requirements. Unlike traditional plandriven methodologies, Agile promotes flexibility, constant stakeholder cooperation, and incremental value delivery (Rigby, et al., 2018). Among Agile frameworks, Scrum is widely embraced due to its structured yet adaptable approach to managing complicated projects (Schwaber & Sutherland, 2020).

In the FarmaY project, the adoption of Scrum was particularly significant given the changing regulatory environment and evolving inventory requirements. The adoption of time-boxed sprints enabled the team to deliver functional increments of the ERP module iteratively, providing early validation of system features. This demonstrates alignment with Agile principles of reducing uncertainty through feedback, although its effectiveness was limited by execution contraints (Project Management Institute, 2021). However, while the structure of Scrum was implemented efficiently, its full benefits were hindered by partial reliance on traditional planning habits, which limited adaptability in practice.

Furthermore, Scrum roles such as the Product Owner and Scrum Master were formed to assist value delivery and process facilitation. While this highlights adherence to Scrum principles, the effectiveness of these roles relies greatly on execution quality. In this scenario, deficiencies in role readiness and team autonomy diminished the projected benefits of Scrum, indicating that framework adoption alone does not ensure success (Moe, et al., 2019).

Overall, although Scrum provided an appropriate structure for managing complexity, its impact was influenced by the organization's level of Agile maturity and capacity to fully embrace its principles.

2.3 Suitability of Agile/SCRUM for the Project

The implementation of Agile Scrum for the FarmaY project exhibits significant alignment between the project environment and the selected development approach. The project operated under conditions of regulatory ambiguity and fluctuating operational requirements, making it unsuitable for standard plan-driven methods. Agile frameworks are specifically built for such situations, as they enable iterative refinement of requirements and stakeholder interaction (Rigby, et al., 2018).

In this situation, Scrum facilitated responsiveness to frequent changes in inventory rules and compliance requirements, which would have been difficult to accommodate using a strict waterfall methodology.

Compared to traditional methodologies, Scrum offers a distinct advantage by enabling incremental delivery of functional components. This allows stakeholders to validate outputs early, particularly in areas like as stock control and expiry management, decreasing the danger of large-scale rework (Project Management Institute, 2021). However, while the system was theoretically well-suited, its performance hinged on rigorous implementation. The team's inclination to revert to traditional planning habits restricted the full realization of Agile benefits, showing partial adoption rather than complete transformation.

Furthermore, Scrum's emphasis on stakeholder collaboration was appropriate given the necessity for close coordination between business and technical teams. Continuous feedback loops increased alignment between system functioning and business needs (Denning, 2018). Nevertheless, this reliance on stakeholder input also required robust governance structures, which were not completely present in the project, leading to concerns such as uncontrolled requirement modifications.

Overall, while Agile Scrum was an appropriate choice for the project, its effectiveness was constrained by execution limitations and organizational readiness.

2.4 Analysis of the Approach taken by the Project Team

The project team chose the Scrum framework to develop the ERP inventory management module, structuring development through iterative sprints and continuous stakeholder involvement. An initial product backlog was built to capture essential functional requirements, which were prioritized based on business impact. This reflects a basic Scrum concept of managing a dynamic and evolving backlog related with changing project demands (Schwaber & Sutherland, 2020). Backlog refinement meant that needs were regularly updated as new insights arose, facilitating adaptive planning in uncertain contexts (Project Management Institute, 2021).

The team created time-boxed sprints to provide functional increments of the system incrementally. This iterative technique permitted early validation of outcomes and allowed stakeholders to submit comments that could be integrated into following rounds. Delivering usable components early boosted stakeholder confidence and reduced project risk, which is an acknowledged advantage of Agile techniques in complex situations (Rigby, et al., 2018). Empirical studies also suggest that iterative delivery boosts responsiveness and speeds value realization in software projects (Denning, 2018).

Daily stand-up meetings were performed to review progress and identify barriers, boosting transparency and collaboration within the team. These short, time-boxed sessions facilitate constant inspection and modification, which are key to Scrum's empirical process control paradigm (Schwaber & Sutherland, 2020). Research demonstrates that such approaches boost communication efficiency and team effectiveness in Agile environments (Moe, et al., 2019). Additionally, sprint reviews enabled stakeholders to verify completed features and suggest improvements, enhancing alignment between system outputs and business needs.

Despite these strengths, numerous drawbacks were discovered in the execution of the technique. The Product Owner, although adequately positioned from a business viewpoint, required further support to interpret technical issues, leading to communication gaps between stakeholders and the development team. This underscores a typical difficulty in Agile initiatives, as effective Product Owners must connect both business and technical domains (Pichler, 2019).

Furthermore, the Scrum Master assumed responsibilities beyond facilitation, including mediating conversations and influencing choices. This overextension demonstrates a lack of fully selforganizing team behavior, as Agile teams are meant to function autonomously with minimum centralized management (Schwaber & Sutherland, 2020). Studies indicate that excessive emphasis on leadership roles might diminish team autonomy and restrict scalability of Agile techniques across businesses (McKinsey & Company, 2018).

The team also displayed a tendency to revert to old planning methodologies, aiming to specify broader system requirements early in the project. This behavior opposes Agile concepts of iterative development and continual refinement (Rigby, et al., 2018). Additionally, dependency on external IT units restricted the team's capacity to act as a fully cross-functional unit, generating delays and reducing responsiveness. Organizational impediments of this sort are commonly identified as important challenges in Agile transformation (Digital.ai, 2023; Project Management Institute, 2021).

Overall, while the project team implemented Scrum methods effectively at a structural level, the approach was hampered by execution issues and organizational limits, indicating a transitional stage of Agile maturity rather than complete adoption.

2.5 Good Practices

A basic strength of the project was the suitable adoption of Agile Scrum for a highly dynamic and requirement-evolving environment. The project contained uncertain and evolving requirements, and

Agile approaches are specifically designed to support such unpredictability through iterative refinement and continual feedback (Rigby, et al., 2018). This connection between project characteristics and approach is widely regarded as a significant success factor in Agile projects. The project further demonstrated effective utilization of Scrum roles, particularly through the appointment of a Scrum Master and a business-oriented Product Owner. The Product Owner provided a guaranteed priority based on business value, which aligns with the principle of value maximization in Scrum (Schwaber & Sutherland, 2020), while the Scrum Master enabled coordination and process adherence. Studies demonstrate that well defined roles considerably boost Agile team effectiveness (Digital.ai, 2023).

Another major strength was the use of iterative development supported by defined Scrum methods, including time-boxed sprints, daily stand-ups, sprint reviews, and retrospectives. These behaviors promote transparency and enable continual inspection and modification, which are key to Agile techniques (Project Management Institute, 2021). Iterative delivery also decreases uncertainty and enhances response to change (Denning, 2018).

A particular noteworthy achievement was early value delivery, with viable results supplied at early phases of the project. Delivering functional increments early helps stakeholders to assess outputs and provide input, which improves alignment with business needs and decreases project risk (Rigby, et al., 2018).

Closely related to this, the project achieved better time and cost efficiency as an effect of Agile implementation. Early validation minimizes rework and optimizes resource use, leading to faster delivery and reduced costs. Industry research demonstrates that Agile approaches contribute to greater productivity and corporate value realization (Digital.ai, 2023), while structured iteration enables better resource control (Project Management Institute, 2021).

The project also displayed significant stakeholder involvement supported by backlog management strategies. Continuous participation through sprint reviews ensured that requirements were confirmed and modified throughout development. Backlog grooming enabled adaptive planning and prioritization, which are critical for handling evolving requirements (Rigby, et al., 2018). Finally, the project benefited from a collaborative and continuously evolving team environment, where cross-functional teamwork and retrospectives facilitated knowledge sharing and process refinement. Continuous improvement is a defining property of Agile teams and contributes to longterm

performance improvements (Moe, et al., 2019), while iterative learning provides sustained organizational competence (Denning, 2018).

2.6 Bad Practices

Despite some benefits, the project demonstrated key limitations that restricted the success of the Agile implementation. A key concern was the gap between good role selection and poor role preparation, notably for the Product Owner. Although the Product Owner was well-positioned from a commercial viewpoint, the lack of basic technical understanding caused communication barriers between stakeholders and the development team. Effective Product Owners are expected to span both commercial and technical domains to ensure efficient decision-making (Pichler, 2019), while insufficient preparation might diminish overall team efficiency (Project Management Institute, 2021).

Another important problem was the overextension of the Scrum Master's position, who acquired responsibilities beyond facilitation, including mediating communication and influencing decisions. This contradicts Scrum principles, where the Scrum Master works as a facilitator rather than a central decision-maker (Schwaber & Sutherland, 2020). Excessive emphasis on leadership roles can cause bottlenecks and hinder the scalability of Agile processes inside businesses. Closely connected to this, the project displayed weak self-organizing team behavior, as team members depended largely on the Scrum Master for guidance and decision-making. Agile teams are supposed to be autonomous and self-managing, and lack of self-organization implies a poor level of Agile maturity (Moe, et al., 2019). This dependency undermines ownership and hinders innovation within the team.

The project also suffered from poor governance of customer feedback and requirement modifications, resulting in scope extension and project delays of roughly 40 days. While Agile emphasizes responsiveness to change, the absence of structured prioritization and impact assessment leads to uncontrolled scope creep. Effective Agile deployment demands balancing flexibility with disciplined backlog management (Rigby, et al., 2018), and failing to do so can significantly affect project deadlines (Project Management Institute, 2021).

Additionally, there was cultural resistance to Agile principles, particularly visible in the team's tendency to resort to traditional planning procedures. Attempts to identify broader system requirements early in the project contradict Agile concepts of iterative development and adaptable

planning. Research reveals that organizational mentality is a significant aspect in successful Agile adoption, and resistance to change might limit its effectiveness (Conforto, et al., 2016; McKinsey & Company, 2018).

Another problem was the lack of comprehensive cross-functional integration due to dependency on external IT units, which generated delays and restricted team autonomy. Scrum implies selfcontained teams capable of delivering end-to-end solutions, whereas external dependencies can drastically limit responsiveness and efficiency (Project Management Institute, 2021).

Finally, the absence of formal control mechanisms such as burndown charts and organized risk management hindered project visibility and proactive decision-making. Without these tools, the team lacked early indicators of performance concerns and dangers, lowering overall management of the project (Schwaber & Sutherland, 2020). Effective Agile teams rely on such techniques to assess progress and handle uncertainty efficiently (Project Management Institute, 2021).

2.7 Lessons Learnt

An important learning from the project is that successful Agile adoption depends on role enablement and organizational readiness rather than the mere execution of the framework. Although Scrum responsibilities were accurately assigned, the lack of adequate technical preparation for the Product Owner generated communication inefficiencies. This emphasizes the need for businesses to engage in capability development to ensure that responsibilities are executed properly, as Agile success relies on both structure and competence (Pichler, 2019).

Another essential lesson is the need to balance flexibility with governance in managing needs and consumer input. While Agile supports adaptation, the project experienced delays owing to unplanned scope changes. Future projects should adopt stronger backlog prioritization and impact assessment procedures, as disciplined execution is vital for retaining efficiency in Agile environments (Rigby, et al., 2018).

The project also highlights that self-organizing teams are crucial for achieving Agile maturity, but they require structured support to develop. The observed dependency on the Scrum Master implies that teams must be actively empowered to take ownership of decision-making. Research suggests that autonomous teams boost performance and innovation in Agile contexts (Moe, et al., 2019), and organizations must build conditions that allow this autonomy.

Another major lesson is the significance of organization-wide alignment in Agile adoption. Dependencies on external units affected the team's responsiveness, suggesting that Agile techniques must extend beyond individual teams to the larger organizational structure (Project Management Institute, 2021).

Finally, the project reinforces that Agile transformation is truly an organizational change undertaking rather than a solely methodological shift, needing alignment of culture, structure, and processes to reap its full benefits (McKinsey & Company, 2018).

2.8 Key Findings

The case study illustrates that Agile Scrum can considerably improve project performance in dynamic and uncertain situations by enabling early value delivery, constant stakeholder interaction, and iterative revision of requirements. The project delivered better delivery speed and operational efficiency compared to traditional ways, supporting evidence that Agile methodologies promote responsiveness, adaptability, and total business value (Rigby, et al., 2018). Iterative delivery also allows stakeholders to review outcomes early, minimizing uncertainty and boosting alignment with company goals.

However, the findings suggest that Agile success depends on execution quality rather than framework adoption alone. Limitations such as inadequate role preparation, over-reliance on the Scrum Master, weak self-organizing teams, and poor governance of requirement modifications show gaps in Agile maturity. These problems represent broader organizational constraints in Agile transitions (McKinsey & Company, 2018).

Overall, Agile offers maximum value when backed by disciplined implementation, empowered teams, and organization-wide alignment (Project Management Institute, 2021)

3. Section 02

3.1 Privacy by Design

Privacy by Design (PbD) integrates privacy protection measures into information system design and architecture from the start, rather than considering privacy as an afterthought. Ann Cavoukian's approach embeds privacy in technology, business, and organizational processes (Cavoukian, 2011).

The General Data Protection Regulation (GDPR) requires companies handling personal data to protect it by design and default (Voigt & von dem Bussche, 2017).
PbD follows seven principles, including proactive measures, default privacy, and data lifecycle protection. These principles minimize data collection, regulate access, and secure data handling. In sensitive systems, PbD increases user trust and minimizes data breach risks (Gürses, et al., 2019; Information Commissioner's Office, 2022).

3.2 Overview of Issues Raised in the Case Study

The FarmaY case demonstrates serious privacy and data protection concerns in the architecture of an ERP inventory management system. Although the system primarily handles operational data such as stock levels, batch numbers, and expiry dates, privacy threats arise through user-related data including access logs and role-based actions. Under GDPR, such information qualifies as personal data and must be processed securely and lawfully (European Union, 2016), demonstrating that privacy concerns are typically underestimated in internal systems and represent a reactive approach to privacy management.
A key challenge is access control and role-based permissions, particularly for functions such as modifying expiry dates. While constraints were placed, their effectiveness depended on proper design and implementation. Weak access controls increase the danger of unauthorized actions, potentially compromising both data integrity and regulatory compliance (National Institute of Standards and Technology, 2020). This demonstrates that access control was viewed as a secondary functional requirement rather than an integrated design priority, weakening overall system governance.
Another key concern is data integrity and auditability, as inconsistencies in expiry records indicate weaknesses in validation systems. This highlights a gap between GDPR requirements for integrity and the system’s implementation, while Privacy by Design principles were not fully embedded at the architectural level (European Union, 2016; Gürses, et al., 2019)

Overall, the case demonstrates that privacy issues originate from poor system design and limited governance integration.

3.3 Challenges of Implementing Privacy by Design

Implementing Privacy by Design (PbD) in enterprise systems such as the FarmaY ERP project presents several practical and organisational challenges. A key difficulty is integrating privacy requirements throughout the early stages of system design. In the case, controls such as access profiles and audit mechanisms appear to have been handled alongside or after functional requirements rather than embedded from the outset. Although PbD requires privacy to be embedded into the system architecture, the project reflects a prioritisation of functionality over privacy integration (Gürses, et al., 2019). This misalignment increases the risk of costly redesigns and non-compliance with GDPR Article 25 (European Union, 2016).

Another problem is balancing data accessibility with data protection. ERP systems require multiple users to access operational data; however excessive access increases the danger of unauthorised exposure. Implementing fine-grained access control is technically challenging and requires ongoing monitoring (National Institute of Standards and Technology, 2020).

Organisational dependence is another limitation. Reliance on external IT teams for implementing access restrictions causes separation between development and governance, restricting effective privacy integration. PbD needs collaboration across technological, legal, and business domains (Information Commissioner's Office, 2022).

Ensuring data integrity and auditability adds complexity, as maintaining accurate records and audit trails requires robust governance. This indicates that GDPR accountability requirements were not fully realised in practice (European Union, 2016).

3.4 Key Findings

The FarmaY case study highlights that privacy issues in enterprise systems extend beyond personal data to encompass access control, data integrity, and system governance. Although the system primarily controls operational data, user access profiles and audit logs add components of personal data processing under GDPR (European Union, 2016). This underscores the need to ensure privacy even in non-customer-facing technologies.

The findings also indicate that Privacy by Design was not fully entrenched at the architectural level, as safeguards such as access limitations and audit procedures were not proactively implemented. This illustrates a widespread gap where functionality is emphasized over privacy, increasing compliance issues (Gürses, et al., 2019).

Overall, successful privacy protection involves early integration of controls, conformity with GDPR principles, and strong governance structures.

4. Conclusion

The study proved that Agile Scrum may dramatically increase project performance in dynamic situations through iterative development, early value delivery, and ongoing stakeholder interaction. However, the findings also indicated that successful implementation relied on effective role execution, team autonomy, and disciplined backlog management rather than framework acceptance alone (Project Management Institute, 2021). In parallel, the Privacy by Design investigation found that corporate systems must include privacy and security measures at the design level, notably in areas like as access control and auditability. Compliance with GDPR principles, including data protection by design, is crucial to maintain system integrity and accountability (European Union, 2016). Overall, both case studies emphasize the need to align methodology, governance, and system design to ensure sustainable and secure project outcomes.

References

Cavoukian, A., 2011. Privacy by Design: The 7 Foundational Principles, Ontario: Information and Privacy Commissioner of Ontario.

Conforto, E. C. et al., 2016. Can Agile Project Management Be Adopted by Industries Other Than Software Development?. Project Management Journal, 47(3), pp. 21 - 34.

Denning, S., 2018. The Age of Agile: How Smart Companies Are Transforming the Way Work Gets Done. New York: AMACOM.

Digital.ai, 2023. 17th State of Agile Report. [Online]
Available at: https://digital.ai/resource-center/analyst-reports/state-of-agile-report/ [Accessed 28th April 2026].

European Union, 2016. General Data Protection Regulation (EU) 2016/679. [Online]
Available at: https://eur-lex.europa.eu/eli/reg/2016/679/oj [Accessed 28th April 2026].

Gürses, S., Troncoso, C. & Diaz, C., 2019. Engineering Privacy by Design. Communications of the ACM, 62(2), pp. 24 - 32.

Information Commissioner's Office, 2022. Guide to Data Protection. [Online]
Available at: https://ico.org.uk/for-organisations/guide-to-data-protection/ [Accessed 28th April 2026].

McKinsey & Company, 2018. The five trademarks of agile organizations. [Online]
Available at: https://www.mckinsey.com/capabilities/people-and-organizational-performance/ourinsights/the-five-trademarks-of-agile-organizations [Accessed 28th April 2026].

Moe, N. B., Dingsøyr, T. & Dybå, T., 2019. Overcoming Barriers to Self-Managing Agile Teams. IEEE Software, 36(2), pp. 20 - 27.

National Institute of Standards and Technology, 2020. NIST Privacy Framework: A Tool for Improving Privacy Through Enterprise Risk Management, s.l.: s.n.

Pichler, R., 2019. Agile Product Management with Scrum. Boston: Addison-Wesley. Project Management Institute, 2021. Agile Practice Guide. Newtown Square, PA: Project Management Institute.

Rigby, D. K., Sutherland, J. & Takeuchi, H., 2018. Embracing Agile. Harvard Business Review, 96(3), pp. 40-50.

Schwaber, K. & Sutherland, J., 2020. The Scrum Guide. [Online]
Available at: https://scrumguides.org/scrum-guide.html [Accessed 28th April 2026].

Voigt, P. & von dem Bussche, A., 2017. The EU General Data Protection Regulation (GDPR): A Practical Guide. Cham: Springer.